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JOMHACK
EMPOWERING INNOVATION

JomCode Coding Bootcamp

CHAPTER 18

HTML and CSS foundations

learn / type / run / explain

Plain meaning and technical view

PLAIN MEANING

HTML describes page meaning. CSS controls how those elements look and fit.

TECHNICAL VIEW

Semantic elements, labels and alt text provide structure. Selectors, box model, flex and grid control presentation.

REAL-WORLD USE

Read React markup and understand the CSS concepts that Tailwind utilities express.

QUICK TIP

Inspect an element in browser developer tools. Tailwind still uses CSS concepts, so learn spacing and layout first.

INSTALL / RUN

No npm package. Open the HTML workbook in a browser. Each panel isolates its own styles.

Worked example 1: Semantic story

chapter_18_html_and_css_foundations.html

```
<main>
  <article>
    <h1>A useful idea</h1>
    <p>By Ada</p>
    <p>Story text belongs in a paragraph.</p>
    <a href="#next">Read another story</a>
  </article>
</main>
```

PREDICT Why use article rather than a div for everything?

Run and explain example 1

EXPECTED OUTPUT

A heading, author, story paragraph and keyboard-focusable link.

WHY IT BEHAVES THIS WAY

Semantic elements describe the content even before styling.

Worked example 2: Labeled input and image

chapter_18_html_and_css_foundations.html

```
<form>
  <label for="title">Title</label>
  <input id="title" name="title" required maxlength="160">
  <button type="submit">Save</button>
</form>
<img src="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg' width='80'
height='40'%3E%3Crect width='80' height='40' fill='skyblue'/%3E%3C/svg%3E"
     alt="Blue demonstration cover" width="80" height="40">
```

PREDICT Does browser validation replace backend validation?

Run and explain example 2

EXPECTED OUTPUT

A labeled required input, Save button and small cover image.

WHY IT BEHAVES THIS WAY

No. Labels improve usability and native constraints help users, but the server must validate too.

Exercise 1: Story page

YOUR TASK

Build a semantic story page with title, author, body and labeled search form.
Add a focus style.

CHECK THESE CASES

One h1. Label targets input ID. Link and button reachable with Tab.

HINT AFTER A TRY

Use browser keyboard testing, not just a screenshot.

Worked example 3: Selectors and box model

chapter_18_html_and_css_foundations.html

```
<style>
  * { box-sizing: border-box; }
  .card { width: 240px; padding: 16px; border: 2px solid #0891b2; margin: 16px; }
  .card h2 { color: #155e75; }
</style>
<article class="card"><h2>Hello</h2><p>Padding is inside the border.</p></article>
```

PREDICT How does border-box change the width calculation?

Run and explain example 3

EXPECTED OUTPUT

A 240px-wide bordered card including its padding and border.

WHY IT BEHAVES THIS WAY

The declared width includes padding and border. Margin remains outside.

Worked example 4: Responsive layout

chapter_18_html_and_css_foundations.html

```
<style>
  .grid { display: grid; gap: 12px; grid-template-columns: 1fr; }
  .item { padding: 16px; background: #e0f2fe; }
  @media (min-width: 600px) { .grid { grid-template-columns: repeat(2, 1fr); } }
</style>
<section class="grid"><article class="item">Story A</article><article class="item">Story
B</article></section>
```

PREDICT What happens when the viewport narrows?

Run and explain example 4

EXPECTED OUTPUT

One column below 600px, two columns at 600px and above.

WHY IT BEHAVES THIS WAY

The base rule remains one column. The media query adds the wider layout only at its threshold.

Exercise 2: Responsive card layout

YOUR TASK

Build two story cards with border-box, consistent spacing and a responsive grid.

CHECK THESE CASES

One column on narrow view. Two above 600px. No horizontal overflow.

HINT AFTER A TRY

Resize the browser and inspect computed spacing.

Checkpoint before moving on

SHOW AND EXPLAIN

Run all four examples, complete both tasks, and explain read react markup and understand the css concepts that tailwind utilities express.

TRANSFER TO POSTLIGHT

Read React markup and understand the CSS concepts that Tailwind utilities express.